

AIM: To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

Objective:

To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.

Procedure:

1. Define the V2X communication parameters such as vehicle speed, transmission range, packet size, and simulation time.
2. Generate different vehicle movement scenarios by varying the vehicle speed.
3. Simulate wireless communication between vehicles using a V2X communication model.
4. Calculate the End-to-End Latency (ms) for data transmission between communicating vehicles.
5. Compute the Packet Delivery Ratio (PDR %) by comparing the number of successfully received packets with transmitted packets.
6. Analyze the effect of increasing Vehicle Speed (km/h) on communication latency and packet delivery performance.
7. Plot MATLAB graphs for End-to-End Latency, Packet Delivery Ratio, and Vehicle Speed.
8. Compare the obtained results to evaluate the efficiency and reliability of the V2X communication system under different traffic conditions.

Matlab Code:

```
clc;

clear;

close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

% Simulated End-to-End Latency (ms)

latency = [18 16 14 12 10 9];

disp('Vehicle Speed (km/h) End-to-End Latency (ms)')

disp([speed' latency'])
```

figure

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(speed,latency,'-o','LineWidth',2)
```

```
title('Vehicle Speed vs End-to-End Latency')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(1:length(latency),latency,'-s','LineWidth',2)
```

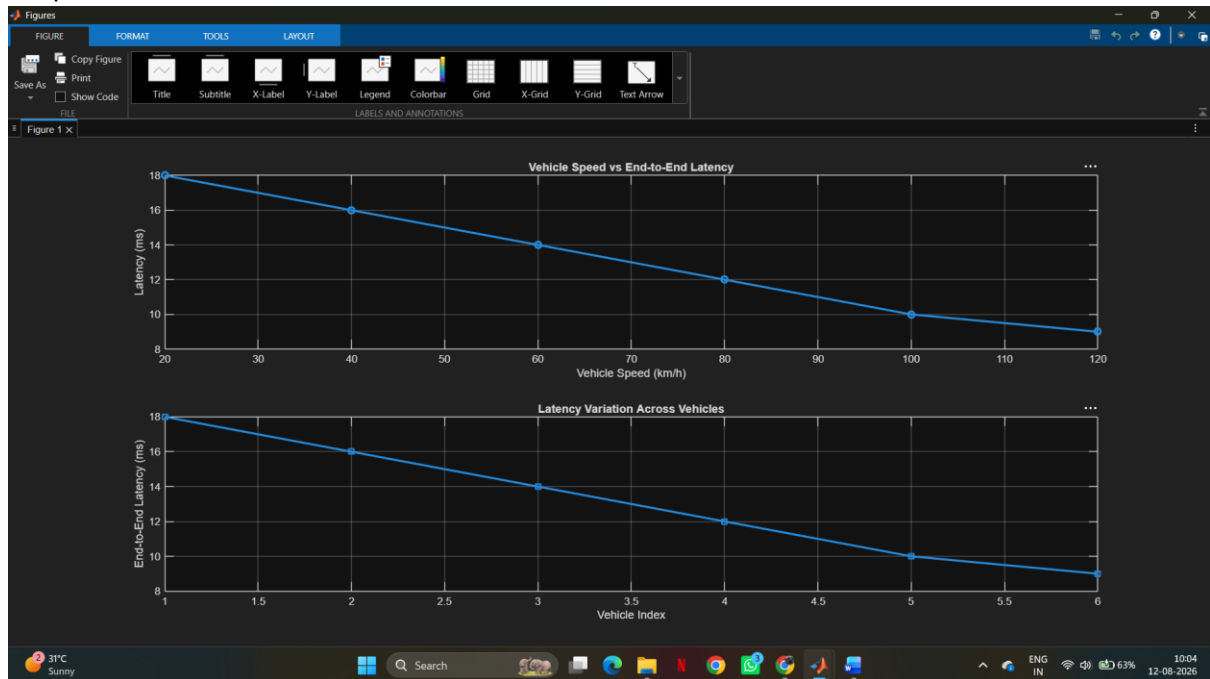
```
title('Latency Variation Across Vehicles')
```

```
xlabel('Vehicle Index')
```

```
ylabel('End-to-End Latency (ms)')
```

```
grid on
```

Output:



Result:

- 1.The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.
- 2.The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.
- 3.The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.